

S.NO	Details of the Paper	Problem Solved	Proposed Solution	Results Achieved	Merits	Demerits	Future Work	Tools / Platform	Performance Metrics
1.	Kavalci, E., Hartshorn, A. Improving clinical trial design using interpretable machine learning based prediction of early trial termination. <i>Sci Rep</i> 13, 121 (2023). https://doi.org/10.1038/s41598-023-27416-7	Forecasting potential delays in clinical trials using data-driven models.	Used XGBoost and SHAP explainability on structured trial data to identify delay factors.	Achieved moderate prediction accuracy; SHAP revealed top delay drivers (phase, site count, sponsor).	Explainable model; helps identify key delay causes.	Ignores unstructured text; predicts only delays, not completion; non-interactive.	Extend model to predict enrollment/ completion success.	Python, scikit-learn, SHAP.	Accuracy, F1-score (exact values not reported).
2.	Teodoro, D., Naderi, N., Yazdani, A. <i>et al.</i> A scoping review of artificial intelligence applications in clinical trial risk assessment. <i>npj Digit. Med.</i> 8, 486 (2025). https://doi.org/10.1038/s41746-025-01886-7	Predicting success/failure of trials using multimodal (molecule + sponsor) data.	Developed Graph Neural Network integrating molecular, sponsor, and drug data.	Achieved strong ROC-AUC (≈ 0.78); improved over logistic baselines.	Multimodal integration, strong performance.	Hard to interpret; complex GNN architecture; no user interface.	Improve explainability and real-world usability.	Python, PyTorch Geometric.	ROC-AUC ≈ 0.78 , F1 ≈ 0.71

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3.	Tianfan Fu, Kexin Huang, Cao Xiao, Lucas M. Glass, Jimeng Sun, HINT: Hierarchical interaction network for clinical-trial-outcome predictions, Patterns, Volume 3, Issue 4, 2022, 100445, ISSN 2666-3899, https://doi.org/10.1016/j.patter.2022.100445 . (https://www.sciencedirect.com/science/article/pii/S2666389922000186)	Predicting enrollment success of clinical trials before execution.	Combined Deep & Cross Network (DCN) with LLM-augmented text features and hierarchical attention.	PR-AUC 0.7002, ROC-AUC 0.7352, outperforming baseline ML models.	First large AI-ready dataset (31k trials); interpretable through attention and SHAP.	Focus only on enrollment, not completion; structured input only.	Extend to predict completion success and dropout rates.	Python, PyTorch, BioBERT, Mistral LLM.	PR-AUC 0.7002, ROC-AUC 0.7352
4.	Wang S, Šuster S, Baldwin T, Verspoor K Predicting Publication of Clinical Trials Using Structured and Unstructured Data: Model Development and Validation Study J Med Internet Res 2022;24(12):e38859 URL: https://www.jmir.org/2022/12/e38859	Estimating enrollment duration and dropout rates.	Used regression models (Random Forest, Linear Regression) on structured variables (phase, sponsor, country).	Predictive accuracy moderate; identified top influencing factors (phase, sponsor).	Simple, interpretable; uses accessible public data.	Ignores textual eligibility data; not deep-learning-based.	Integrate unstructured data and advanced deep models.	R, Python (scikit-learn).	RMSE, R ² (exact values not reported).

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5.	Yue Wang, Tianfan Fu, Yinlong Xu, Zihan Ma, Hongxia Xu, Bang Du, Yingzhou Lu, Honghao Gao, Jian Wu, and Jintai Chen. 2024. TWIN-GPT: Digital Twins for Clinical Trials via Large Language Model. ACM Trans. Multimedia Comput. Commun. Appl. Just Accepted (July 2024). https://doi.org/10.1145/3674838	Simulating patient behavior and outcomes using LLM-based digital twins.	Fine-tuned GPT model to generate synthetic patient visit histories and simulate trial outcomes.	Demonstrated realistic patient-level data generation for simulation testing.	Innovative use of LLMs; enables trial simulation.	Heavy computation; not interpretable or deployable for small setups.	Develop lighter, interpretable LLM models for prediction.	Python, PyTorch, GPT fine-tuning.	No numeric metrics (qualitative validation).
6.	Zifeng Wang, Cao Xiao, and Jimeng Sun. 2023. SPOT: Sequential Predictive Modeling of Clinical Trial Outcome with Meta-Learning. Proceedings of the 14th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics (BCB '23). Association for Computing Machinery, New York, NY, USA, Article 53, 1–11. https://doi.org/10.1145/3584371.3613001	Automated reasoning across trial sub-tasks (enrollment, efficacy, safety).	Multi-agent framework using LLMs for cooperative reasoning.	Conceptually effective in decomposing complex trial tasks.	Novel reasoning approach across multiple LLM agents.	Conceptual only; no real-world validation or performance metrics.	Build single-agent practical implementations; validate with data.	Python, LangChain, LLM APIs.	Not reported (conceptual work).

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7	Markey, N., Howitt, B., El-Mansouri, I. <i>et al.</i> Clinical trials are becoming more complex: a machine learning analysis of data from over 16,000 trials. <i>Sci Rep</i> 14, 3514 (2024). https://doi.org/10.1038/s41598-024-53211-z	Clinical trials are increasingly complex, with growing numbers of endpoints, eligibility criteria, and treatment arms — but there was no systematic, data-driven way to measure or predict this complexity.	Applied ML-based regression and classification techniques using features such as the number of eligibility criteria, interventions, endpoints, sponsors, therapeutic area, and trial phase to estimate complexity levels. Temporal trends were also analyzed to evaluate how complexity has changed over decades.	Found a steady increase in trial complexity over the past 15 years, especially in oncology and late-phase industry-sponsored trials. The ML models accurately predicted high-complexity trials and identified eligibility count and number of endpoints as top predictors.	Provides a clear, quantitative framework for measuring trial complexity. Large dataset improves generalizability and robustness. Identifies critical drivers of trial complexity growth.	Focuses on complexity, not completion or success outcomes. Uses primarily structural features; lacks text-based semantic insights. Complexity score is a proxy — may not directly relate to operational outcomes.	.Integrate textual protocol analysis (using NLP/LLMs) for semantic understanding; link trial complexity to completion rate and cost to enhance predictive power.	Python; scikit-learn and LightGBM for ML modeling; data sourced from ClinicalTrials.gov.	AUC $\approx$ 0.79 for complexity classification; $R^2 \approx$ 0.71 for regression-based complexity estimation
8	Chen D, Cao C, Kloosterman R, Parsa R, Raman S Trial Factors Associated With Completion of Clinical Trials Evaluating AI: Retrospective Case-Control Study <i>J Med Internet Res</i> 2024;26:e58578 URL: https://www.jmir.org/2024/1/e58578 DOI: 10.2196/58578	Many AI-related clinical trials end early or stall, but there is limited understanding of which characteristics influence completion.	Used logistic regression and Random Forest models on ClinicalTrials.gov data (2010–2023) to evaluate completion likelihood based on features like sponsor type, phase, number of sites, enrollment size, and study design.	Industry-funded and multi-site trials were more likely to complete successfully. Academic and early-phase studies had higher termination rates. Models achieved AUC $\approx$ 0.82.	Directly investigates trial completion, aligning with ClinWise objectives. Provides practical design insights for improving success rates. Combines interpretability and predictive accuracy.	Limited to AI-related trials (small dataset). No analysis of text-based or patient-level variables. Static model—no dynamic progress tracking.	Include NLP features from eligibility text, expand beyond AI studies, and develop real-time monitoring systems.	Python, scikit-learn; ClinicalTrials.gov dataset	AUC $\approx$ 0.82, Accuracy $\approx$ 0.77, F1-score, feature importance ranking.

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9	Kavalci, E., Hartshorn, A. Improving clinical trial design using interpretable machine learning based prediction of early trial termination. <i>Sci Rep</i> 13, 121 (2023). https://doi.org/10.1038/s41598-023-27416-7	Many clinical trials terminate early due to design or operational issues, causing financial and scientific loss. There was no transparent, data-driven model to predict such early terminations before they happen..	Developed ensemble ML classifiers (Random Forest, XGBoost, Logistic Regression) using features like study phase, sponsor type, intervention count, enrolment size, and therapeutic area. Applied SHAP (Shapley Additive Explanations) for interpretability to show which factors most influence termination.	Achieved an AUC of ≈ 0.80 in classifying “terminated” vs “completed” trials. Key predictors identified included small sample sizes, industry-funded trials, later phases, and high complexity.	Transparent explainability using SHAP. Robust performance across multiple ML algorithms. Provides actionable insights for improving trial design and planning.	Uses only structured tabular data; ignores textual or semantic data (e.g., eligibility criteria). Predicts termination rather than successful completion. Static model—can not handle continuous updates during trial execution.	Expand to predict trial <i>completion</i> probability (the opposite of termination). Integrate textual embeddings from protocols or eligibility text. Implement dynamic, real-time monitoring of trial status.	Python 3.x; scikit-learn; XGBoost; SHAP library; data from ClinicalTrials.gov.	AUC ≈ 0.80 ; Accuracy; Precision–Recall AUC; feature importance via SHAP values.
10	Chen, J., Hu, Y., Cai, M. et al. TrialBench: Multi-Modal AI-Ready Datasets for Clinical Trial Prediction. <i>Sci Data</i> 12, 1564 (2025). https://doi.org/10.1038/s41597-025-05680-8	Existing clinical trial datasets are fragmented, inconsistent, and lack standardized labeling, making it difficult to train or benchmark AI models effectively. This limits reproducibility and model comparison across studies.	Developed TrialBench , integrating structured, textual, and molecular data from ClinicalTrials.gov and other biomedical sources. Curated clean, labeled datasets for eight key AI-solvable trial prediction tasks , ensuring consistent formatting, metadata normalization, and public accessibility for researchers.	TrialBench established the first standardized benchmark for AI in clinical trial analysis, with 24 curated datasets . Demonstrated baseline performances for common ML and deep learning models across all tasks, providing reference metrics for future models.	.First unified multi-modal dataset for clinical trial AI research. Covers a wide range of predictive tasks (enrollment, duration, dropout, etc.). Enables fair benchmarking and reproducibility across models and studies.	.Benchmark datasets still rely mainly on ClinicalTrials.gov (may lack diversity). Focused on dataset preparation, not advanced modeling. Lacks real-time trial progress or site-level data.	Add real-time data streams and patient-level updates; integrate LLM-derived text embeddings; expand to device and behavioral trials.	Python; Data pipelines built using Pandas, PyTorch, Hugging Face for text preprocessing; datasets hosted on open scientific data repositories.	Baseline metrics (AUC, PR-AUC, F1) reported for benchmark models per task; specific task metrics depend on prediction type (classification/regression).

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11	Cho J, Xu Q, Wong CH, Lo AW. Predicting clinical trial duration via statistical and machine learning models. Contemp Clin Trials Commun. 2025 Mar 31;45:101473. doi: 10.1016/j.conctc.2025.101473. PMID: 40248172; PMCID: PMC12005917.	Clinical trial duration is a major bottleneck in drug development, yet few predictive frameworks exist to estimate how long a trial will take before it finishes or is terminated. Existing linear models fail to capture the non-linear dependencies between design, phase, and operational variables.	Combined Cox Proportional Hazards, DeepSurv, and ensemble ML models (Random Forest, Gradient Boosted Trees). These models learned from structured trial features such as sponsor type, number of arms, disease area, and eligibility count to estimate time-to-completion.	DeepSurv achieved the highest predictive performance, accurately estimating duration trends and handling censored data better than traditional methods. Key predictors included trial phase, sponsor category, and patient enrollment size.	Integrates time-to-event modeling with ML. Can handle censored or ongoing trials effectively. Offers interpretability via feature importance analysis.	Ignores text-based features (eligibility criteria semantics). High computational cost for neural survival models. Relies on complete timestamp data; missing data reduced usable trials.	Add textual and real-time operational data to improve accuracy; extend to domain-specific (oncology, cardiovascular) trials for validation.	Python, TensorFlow/Keras (DeepSurv), scikit-survival, lifelines library.	Concordance Index (C-index), Mean Absolute Error (MAE), Log-rank test.
12	Siqi Zhang et al. Enrollment Success, Factors, and Prediction Models in Cancer Trials (2008-2019). <i>JCO Oncol Pract</i> 19, 1058-1068(2023). DOI: 10.1200/OP.23.00147	Poor enrollment rates are a leading cause of delays or termination in oncology trials. Despite its importance, there is limited quantitative understanding of which design or demographic factors predict enrollment success.	Applied logistic regression and gradient boosting algorithms to classify trials as "successful" (≥80% target accrual) vs. "unsuccessful." Input features included disease type, sponsor, phase, site count, and geographic scope.	Achieved AUC ≈ 0.78, showing good discrimination between successful and failed trials. Found that multi-center, industry-funded trials with broader eligibility had higher recruitment success.	Domain-specific insight for oncology. Balanced interpretability (logistic regression) and accuracy (boosted trees). Provides actionable recruitment recommendations.	Limited to oncology; generalization to other diseases uncertain. Structured data only—no text or patient-level details. Static prediction; lacks time-dynamic updates.	Add patient and site-level recruitment data; incorporate text-derived eligibility embeddings; extend to predict dropout risk.	Python (scikit-learn, XGBoost); data sourced from ClinicalTrials.gov and ASCO registries.	.AUC ≈ 0.78, Accuracy, F1-score, and Odds Ratios for feature significance.

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13	Aliper, A., Kudrin, R., Polykovskiy, D., Kamy, P., Tutubalina, E., Chen, S., Ren, F. and Zhavoronkov, A. (2023), Prediction of Clinical Trials Outcomes Based on Target Choice and Clinical Trial Design with Multi-Modal Artificial Intelligence. Clin Pharmacol Ther, 114: 972-980. https://doi.org/10.1002/cpt.3008	Early-stage trial failures often stem from poor target selection and limited predictive insight into whether a molecule is likely to succeed in later phases. Traditional statistical models lack the ability to integrate complex biological and molecular relationships.	Designed a transformer-based predictive framework that encodes molecular descriptors, target-disease associations, and biological pathway data. The model learns high-level patterns across drugs and diseases to predict whether Phase II trials will succeed or fail.	The model achieved AUC > 0.80 in internal validation and was successfully integrated into a commercial prediction system (inClinico). Demonstrated capability to forecast outcomes months before public trial readouts.	First transformer-based approach for Phase II outcome prediction. Integrates drug, target, and disease-level data for molecular context. Provides early-stage decision support for pharma R&D.	Focuses on molecular data, not operational or enrollment aspects. Model complexity reduces interpretability for clinicians. Limited generalization beyond drug-related trials.	Incorporate operational and patient-level variables (e.g., site activity, eligibility data). Extend transformer models for multimodal (molecular + text) trial understanding.	Python, PyTorch; transformer neural architecture; proprietary dataset + public biomedical sources (DrugBank, PubChem).	AUC > 0.80; Accuracy; Precision–Recall balance for classification.
14	Long, B.; Lai, S.-W.; Wu, J.; Bellur, S. Predicting Phase 1 Lymphoma Clinical Trial Durations Using Machine Learning: An In-Depth Analysis and Broad Application Insights. Clin. Pract. 2024, 14, 69-88. https://doi.org/10.3390/clinpract140007	Phase I oncology trials, especially in lymphoma, often experience time overruns due to recruitment and logistical factors. Predicting such delays early can save cost and optimize planning.	Used supervised ML classifiers (Random Forest, Gradient Boosting, Logistic Regression) trained on trial-level metadata (sample size, phase subtype, region, sponsor, site count, disease subtype) to classify trials as “on-time” vs. “delayed.”	Random Forest achieved Accuracy ≈ 81% and AUC ≈ 0.83 ; identified key predictors including sponsor type, multi-site involvement, and eligibility stringency.	First focused model for lymphoma-specific duration prediction. Clear identification of operational delay factors. Provides interpretable feature importance.	Limited dataset size (only lymphoma trials). No time-to-event modelling — treated duration as binary. Ignores text data and patient-level variability.	Incorporate natural language features from eligibility criteria; expand to other cancer types; transition to survival-based duration modeling.	Python, scikit-learn, XGBoost; ClinicalTrials.gov data extraction.	Accuracy ≈ 81%, AUC ≈ 0.83, F1-score, and feature importance ranking.

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15	Teodoro, D., Naderi, N., Yazdani, A. <i>et al.</i> A scoping review of artificial intelligence applications in clinical trial risk assessment. <i>npj Digit. Med.</i> 8, 486 (2025). https://doi.org/10.1038/s41746-025-01886-7	Although AI has been widely adopted in healthcare, there was no consolidated overview of how it is used for risk prediction and management in clinical trials, leading to fragmented progress.	Reviewed and categorized studies by risk type, data used, and ML methods. Analyzed algorithmic trends and performance metrics.	Found most studies used Random Forest, SVM, and XGBoost ; newer ones used transformers and GNNs . Reported high AUROC scores (up to ~0.96).	First systematic mapping of AI in trial risk prediction. Highlights algorithmic and data trends.	Review only; no new model or dataset. Inconsistent metrics across studies.	Develop prospective, real-time AI models for risk prediction; standardize evaluation metrics and benchmarks; explore multi-task learning combining multiple risk types.	Databases mining (PubMed, Web of Science); manual review; analysis in Excel/Python.	Most studies used AUROC, Accuracy, or F1-score; top models achieved AUROC values near 0.95–0.96.
16	Miller MI, Shih LC, Kolachalama VB. Machine Learning in Clinical Trials: A Primer with Applications to Neurology. <i>Neurotherapeutics</i> . 2023 Jul;20(4):1066-1080. doi: 10.1007/s13311-023-01384-2. Epub 2023 May 30. PMID: 37249836; PMCID: PMC10228463.	Clinical trials face inefficiencies in design, recruitment, and monitoring. There's a need to understand where ML can enhance these processes.	Reviewed supervised, unsupervised, and deep learning applications for tasks such as patient selection, enrollment prediction, endpoint modeling, and safety monitoring .	No new model proposed; provides structured insights and practical guidance for implementing ML tools in trial workflows.	Comprehensive overview for new researchers. Highlights best practices and pitfalls in ML-based trial design.	Conceptual; lacks empirical validation. Focused mainly on neurological trials.	Encourages prospective validation, use of unstructured text data, and emphasis on explainable AI.	General overview (Python, ML frameworks discussed).	Not applicable (review-based).